Business Requirements Document (BRD)

NovaTech Industries — Intelligent Procurement & Logistics Management Platform

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Related documents	business-case.md; supplier-evaluation-requirements.md; supplier-score-normalization-validation.md; supplier-evaluation-data-dictionary.md

Case-study notice: NovaTech Industries is a fictional company. All business figures, roles, and operational scenarios are simulated for training and portfolio development.

1. Executive Summary

NovaTech Industries requires a centralized procurement and logistics management platform to reduce manual work, standardize purchasing decisions, improve inventory visibility, and provide reliable operational reporting. The solution will integrate supplier records, product catalogs, purchase requisitions, supplier quotations, purchase orders, warehouse inventory, shipment tracking, business intelligence, and controlled automation. AI assistance will be introduced after deterministic business rules and core data processes are validated.

This BRD defines **what** the business needs, **who** uses the system, and **how success will be accepted**. Technical implementation choices belong in the system architecture document.

2. Business Context and Objectives

NovaTech operates a simulated network of 50 suppliers, 200 SKUs, and three warehouses. Procurement, logistics, and inventory information is currently fragmented across spreadsheets and manual communications.

2.1 Business objectives

ID	Objective	Proposed measurement
BO-001	Reduce procurement cycle time	Average requisition-to-order processing time
BO-002	Improve supplier delivery performance	On-time delivery rate
BO-003	Reduce administrative effort	Manual minutes per purchasing transaction
BO-004	Improve inventory availability	Stockout incidents per quarter
BO-005	Improve quotation evaluation efficiency	Minutes per quotation evaluation
BO-006	Increase purchasing data accuracy	Percentage of purchase orders passing validation

2.2 Illustrative baseline and targets

KPI	Simulated baseline	Proposed target
Requisition processing time	3.5 days	1.5 days
On-time supplier deliveries	87%	95%
Manual procurement processing time	100% baseline	40% reduction
Stockout incidents	24 per quarter	12 per quarter
Quotation evaluation time	90 minutes	20 minutes
Purchase order data accuracy	93%	99%

These values are assumptions for the case study, **not observed company results**. Targets will be assessed against synthetic scenarios during testing.

3. Scope

3.1 In scope

• Supplier master data, approval status, product offerings, and supplier performance.

• Product catalog and purchasing information.

• Purchase requisitions, quotation capture and comparison, purchase order creation, and approval controls.

• Inventory balances, stock movements, and reorder alerts.

• Shipment tracking, milestones, exceptions, and delivery confirmation.

• Management dashboards and KPI calculations.

• Workflow automation and AI-assisted recommendations with human oversight.

• Synthetic data generation, validation, and local demonstration.

3.2 Out of scope for initial release

• Real supplier or carrier integrations and live commercial transactions.

• Payments, invoicing, tax filing, and full accounting ledger.

• Enterprise ERP replacement.

• Autonomous purchase order approval or dispatch by AI.

• Production-grade multi-tenant hosting and external customer access.

These may be evaluated in later phases, but are not prerequisites for the first working release.

4. Stakeholders and User Roles

Role	Primary responsibilities	Required access
Procurement Officer	Maintain sourcing activity, requisitions, quotations, and orders	Suppliers, products, procurement
Procurement Manager	Review exceptions, approve orders, oversee supplier performance	Procurement, approvals, reports
Warehouse Operator	Receive goods and record stock movements	Warehouse, inventory

Role	Primary responsibilities	Required access
Logistics Coordinator	Update shipment status and investigate delays	Shipments, delivery events
Operations Manager	Review operational performance and trends	Dashboards and reporting
System Administrator	Maintain user access and business configurations	Administrative settings

Permissions will follow the principle of least privilege. A user should access only the capabilities needed for their role.

5. End-to-End Business Process

1. **Identify demand:** An inventory threshold or authorized user creates a purchase requirement.
2. **Create requisition:** Capture product, quantity, requested delivery date, requester, and justification.
3. **Request and register quotations:** Obtain offers from relevant approved suppliers.
4. **Validate eligibility:** Check supplier status, product approval, quotation validity, quantities, delivery constraints, and compliance.
5. **Evaluate offers:** Apply weighted criteria and present a ranked, explainable recommendation.
6. **Review and approve:** An authorized person confirms the sourcing decision and purchase order.
7. **Manage shipment:** Track dispatch, transit, arrival, and exceptions.
8. **Receive goods:** Record accepted quantities and inventory movements.
9. **Monitor performance:** Update supplier metrics and management dashboards.

6. Functional Requirements

Priority uses MoSCoW: **Must** = essential core capability; **Should** = important later capability; **Could** = optional enhancement.

ID	Requirement	Priority	Acceptance summary
FR-001	Create, view, and update supplier records	Must	Supplier IDs unique; required fields validated; changes traceable
FR-002	Maintain product catalog and supplier-product offers	Must	Product identifiers unique; supplier-product relationships valid
FR-003	Create and manage purchase requisitions	Must	Required fields validated; status changes controlled
FR-004	Register and compare supplier quotations	Must	Offers linked to requisitions and products; expiry checked
FR-005	Recommend suppliers using weighted scoring	Must	Eligibility applied first; scores explainable and reproducible
FR-006	Generate purchase orders from approved requests	Must	Order linked to supplier and source; unauthorized creation blocked
FR-007	Maintain inventory balances and movements	Must	Receipts update stock through auditable movements
FR-008	Track shipments and delivery events	Must	Milestones recorded; late shipments identifiable
FR-009	Calculate procurement and logistics KPIs	Must	Metrics based on defined formulas and traceable records
FR-010	Generate reorder and exception alerts	Should	Alerts triggered by configured thresholds

ID	Requirement	Priority	Acceptance summary
FR-011	Support configurable approval workflows	Should	Approval status, actor, and timestamp retained
FR-012	Provide AI procurement assistance	Should	Outputs reference source data; no autonomous approval
FR-013	Extract draft quotation details from documents	Could	Human validation before records are committed
FR-014	Support demand forecasting	Could	Forecasts evaluated against historical synthetic data

6.1 Detailed requirement: FR-005 — Supplier Recommendation

Business owner: Procurement Manager. **Trigger:** A requisition has at least one supplier quotation ready for evaluation.

Required inputs: Product, quantity, required delivery date, supplier status and category approval, quotation prices and currency, quotation validity, lead time, supplier quality history, and delivery reliability history.

Stage A — Mandatory eligibility:

Rule ID	Condition
BR-001	Supplier is active
BR-002	Supplier is approved for the requested product category
BR-003	Quotation is valid on evaluation date
BR-004	Quantity and minimum-order constraints are satisfied
BR-005	Promised delivery meets the required delivery date
BR-006	Applicable quality and compliance requirements are satisfied

Stage B — Weighted scoring of eligible quotations:

Criterion	Weight	Scoring direction
Comparable total purchasing cost	40%	Lower is better
Delivery lead time	25%	Shorter is better
Quality acceptance rate	20%	Higher is better
On-time delivery reliability	15%	Higher is better

The weighted score is $S = 0.40*C + 0.25*D + 0.20*Q + 0.15*R$, where each component is normalized to 0–100. Proposed initial normalization is $C = 100 * minimum_eligible_cost / quotation_cost$, $D = 100 * minimum_eligible_lead_time / quotation_lead_time$, $Q = 100 * accepted_units / inspected_units$, and $R = 100 * on_time_deliveries / completed_deliveries$. Positive denominators and consistent units are required. Zero-day delivery, insufficient performance history, and incomparable cost components require explicit handling before implementation.

FR-005 acceptance criteria:

1. Ineligible offers are excluded and the reason is recorded.

2. All monetary quotations are converted to a comparable evaluation currency using a recorded exchange rate.

3. All four criterion scores fall between 0 and 100.

4. Configured weights sum to 100% before a production evaluation is run.

5. Eligible offers are ranked by descending final score.

6. The system displays underlying criterion scores and reasons for its recommendation.

7. If no eligible quotation exists, the system reports that outcome without inventing a recommendation.

8. Evaluation inputs, quotation set, weights, formulas/version, and timestamp are retained for audit.

9. A human reviewer must approve the purchasing decision before order issuance.

7. Business Data Requirements

The system will require logically separated datasets. The field-level specification is maintained in the supplier evaluation data dictionary and will be extended for inventory and logistics.

Data domain	Key entities	Main purpose
Supplier	Suppliers, supplier-product offers, supplier performance	Identify suppliers and sourcing conditions
Product	Products, categories	Describe purchasable items
Procurement	Requisitions, requisition items, quotations, quotation items	Capture demand and commercial offers
Purchasing	Purchase orders, order items	Record approved commitments
Inventory	Warehouses, inventory balances, inventory movements	Monitor available stock and goods receipts
Logistics	Shipments, shipment events	Track physical delivery
Decision support	Evaluation criteria, evaluation results	Reproduce and explain recommendations

7.1 Core data principles

• Use unique stable identifiers for each business entity.

• Enforce referential integrity across related datasets.

• Keep monetary amount and currency together; retain exchange-rate source and effective date.

• Record transaction timestamps and statuses consistently.

• Distinguish missing information from measured zero values.

• Preserve source records and decision history rather than overwriting audit evidence.

• Generate synthetic data with realistic business relationships and controlled exceptions.

8. Non-Functional Requirements

ID	Requirement	Proposed verification
NFR-001	Data integrity	Invalid foreign keys and duplicate unique IDs are rejected
NFR-002	Traceability	Key transactions and recommendations retain audit details
NFR-003	Security	Role-based permissions for sensitive actions
NFR-004	Reliability	Core workflows handle invalid or incomplete inputs without silent failure
NFR-005	Maintainability	Modular components, tests, and documented interfaces

ID	Requirement	Proposed verification
NFR-006	Usability	Users can understand statuses, errors, and recommendation rationale
NFR-007	Local operability	Initial demonstrator runs on the project's local development environment
NFR-008	Reproducibility	Synthetic data can be regenerated from documented seeds and parameters

Numeric performance thresholds will be agreed during technical design and testing, rather than invented in this BRD.

9. Business Rules and Controls

• Purchase orders require the appropriate human approval.

• Inventory balances change only through validated stock transactions.

• Shipment receipt quantities cannot silently exceed documented quantities.

• Expired or unapproved quotations cannot be recommended.

• Supplier recommendations are decision support, not autonomous purchasing authority.

• Data quality issues must be visible and must not be concealed by default scores.

• Exceptions and overrides require a reason and an authorized actor.

10. Reporting and KPIs

The initial reporting scope includes procurement spend, quotation processing time, requisition-to-order cycle time, supplier on-time delivery, quality acceptance rate, stockout frequency, open purchase orders, and delayed shipments. Each KPI will require an explicit numerator, denominator, reporting period, and treatment of incomplete records in a separate KPI specification.

11. Assumptions, Dependencies, and Open Decisions

11.1 Assumptions

• The organization and all transactions are fictional.

• The project begins with a small dataset and scales to a larger connected synthetic environment.

• Initial development and demonstration occur locally.

• AI capabilities are introduced after the deterministic workflow is reliable.

11.2 Open decisions

ID	Decision needed	Current position
OD-001	Minimum history required for quality and reliability	To be agreed
OD-002	Missing-history handling and manual-review thresholds	Manual review proposed
OD-003	Same-day / zero-day delivery normalization	To be defined
OD-004	Total landed-cost components and exchange-rate policy	To be defined
OD-005	Approval thresholds by order value and risk	To be defined
OD-006	Supplier scoring tie-break policy	To be defined

ID	Decision needed	Current position
OD-007	Detailed inventory and logistics field dictionaries	Future checkpoint

12. Delivery Phases and Traceability

Phase	Primary scope	Requirements
1	Supplier comparison prototype	FR-001, FR-002, FR-004, FR-005
2	Procurement data and API	FR-003, FR-006
3	Inventory and logistics	FR-007, FR-008
4	Reporting and application UI	FR-009
5	Workflow automation	FR-010, FR-011
6	AI assistance and extensions	FR-012, FR-013, FR-014

Requirements should be traceable to test cases and the datasets used to validate them.

13. Acceptance and Change Control

This document is a **working baseline**, not a signed stakeholder approval. Requirements are ready for implementation only when their acceptance criteria, input fields, exception handling, and business ownership are sufficiently defined. Changes should update the document version and record the affected requirement IDs.

Checkpoint 2.1 outcome

The project has identified business stakeholders, end-to-end process boundaries, the initial functional requirement catalog, non-functional expectations, data domains, and the weighted supplier evaluation strategy. Detailed FR-005 formulas and the first data dictionary are documented separately to avoid duplicating technical specifications.

Next documentation step: Resolve the remaining scoring-policy decisions, then begin the connected synthetic dataset design and generation exercises.